

Artificial Intelligence AGENT & ENVIRONMENT



Summary of Previous Lecture

- Weak and Strong AI
- Acting humanly
- Think like humans
- think rationally
- Acting rationally
- Turing Test
- Chinese Room Argument



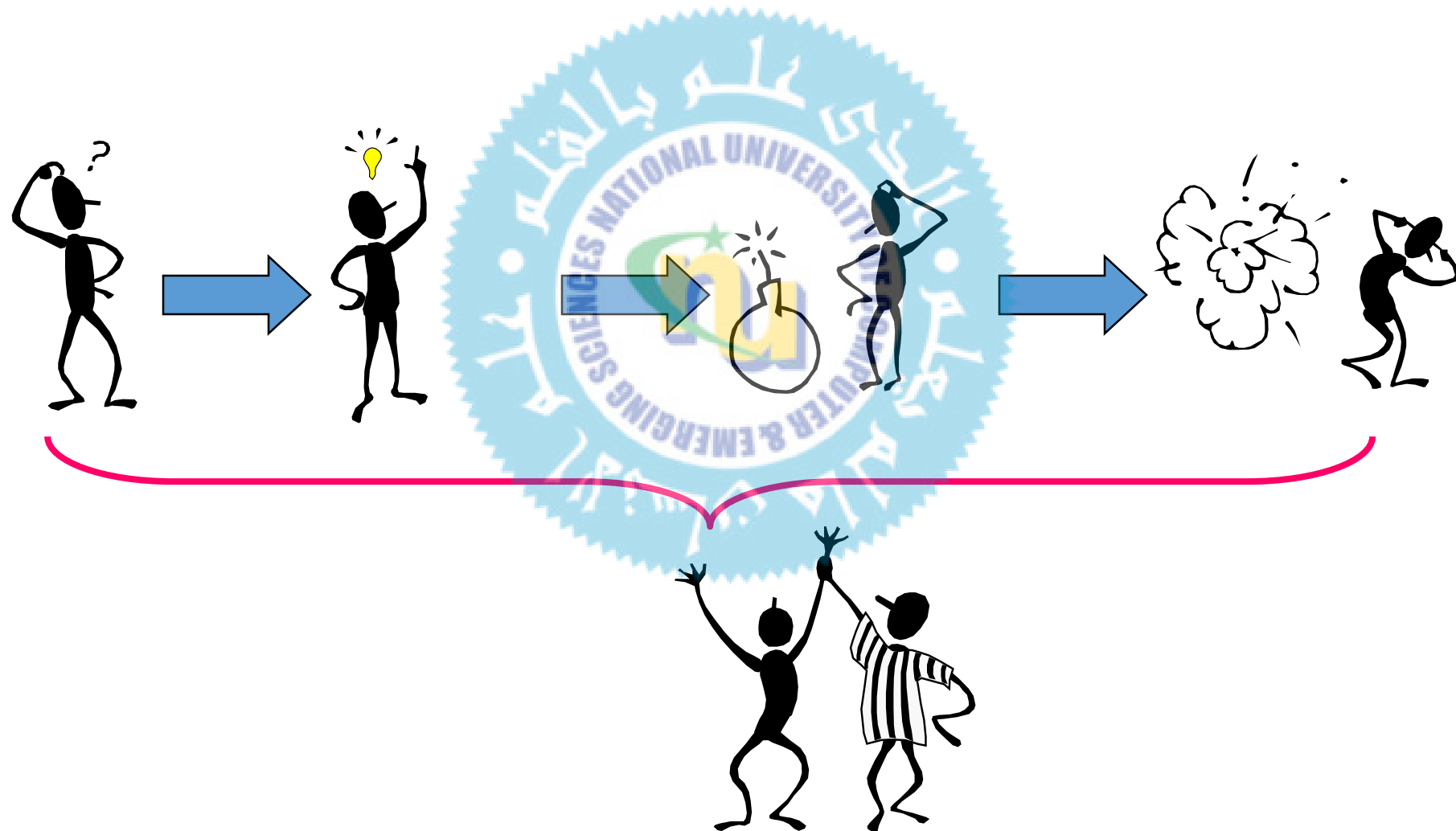
Today's Lecture

- What is an Intelligent agent?
- Agents & Environments
- Performance measure
- Environment
- Actuators
- Sensors
- Features of intelligent agents



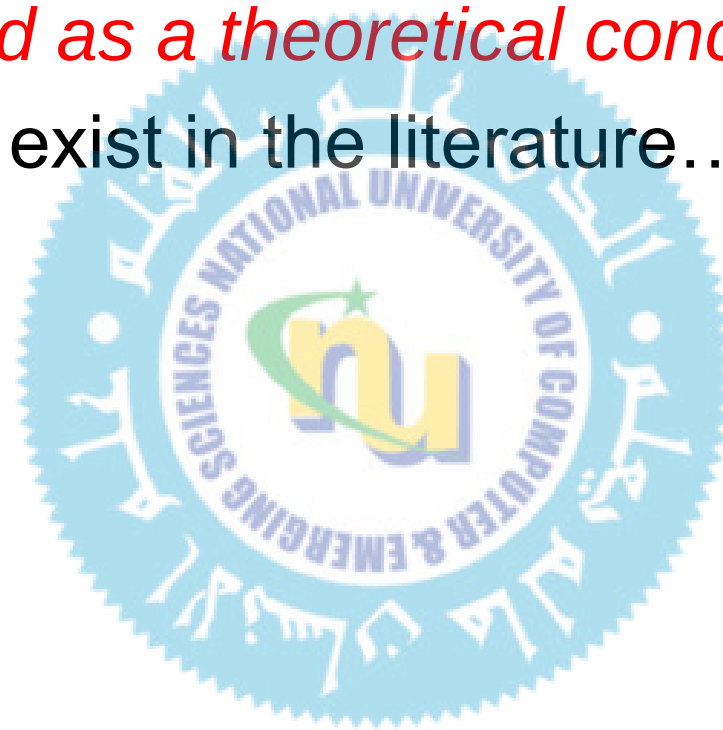
Ability to Exist to be Autonomous, Reactive, Goal-Oriented, etc.

- are the basic abilities of an Intelligent Agent



What is an Intelligent agent?

- “Agent” *can be considered as a theoretical concept from AI.*
- Many different definitions exist in the literature.....

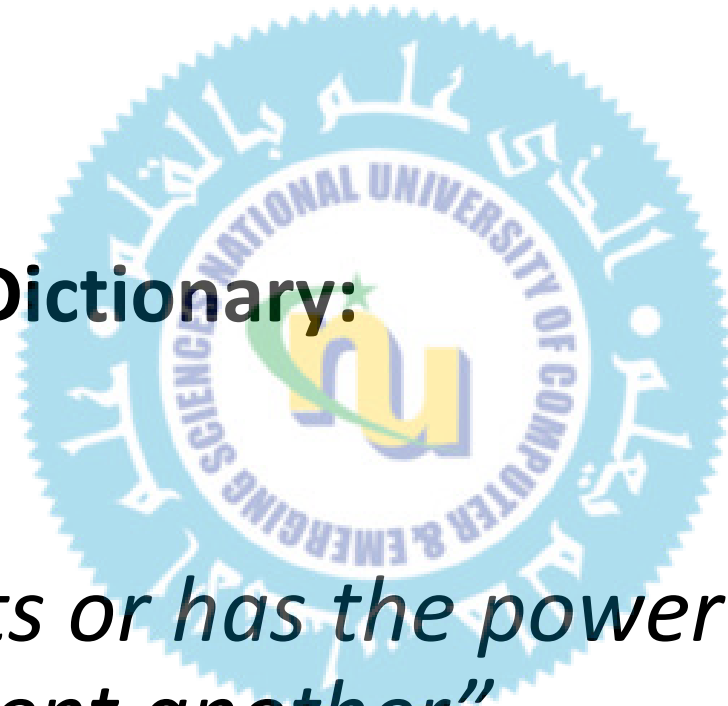


Agent Definition (2)

American Heritage Dictionary:

agent -

“... one that acts or has the power or authority to act... or represent another”

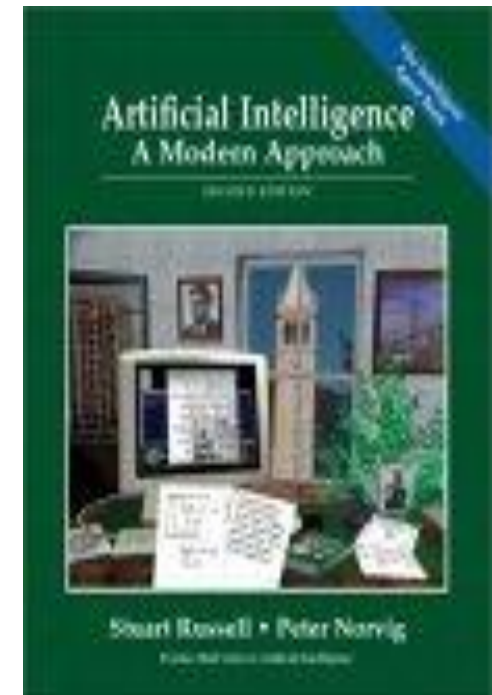


Agent Definition (3)

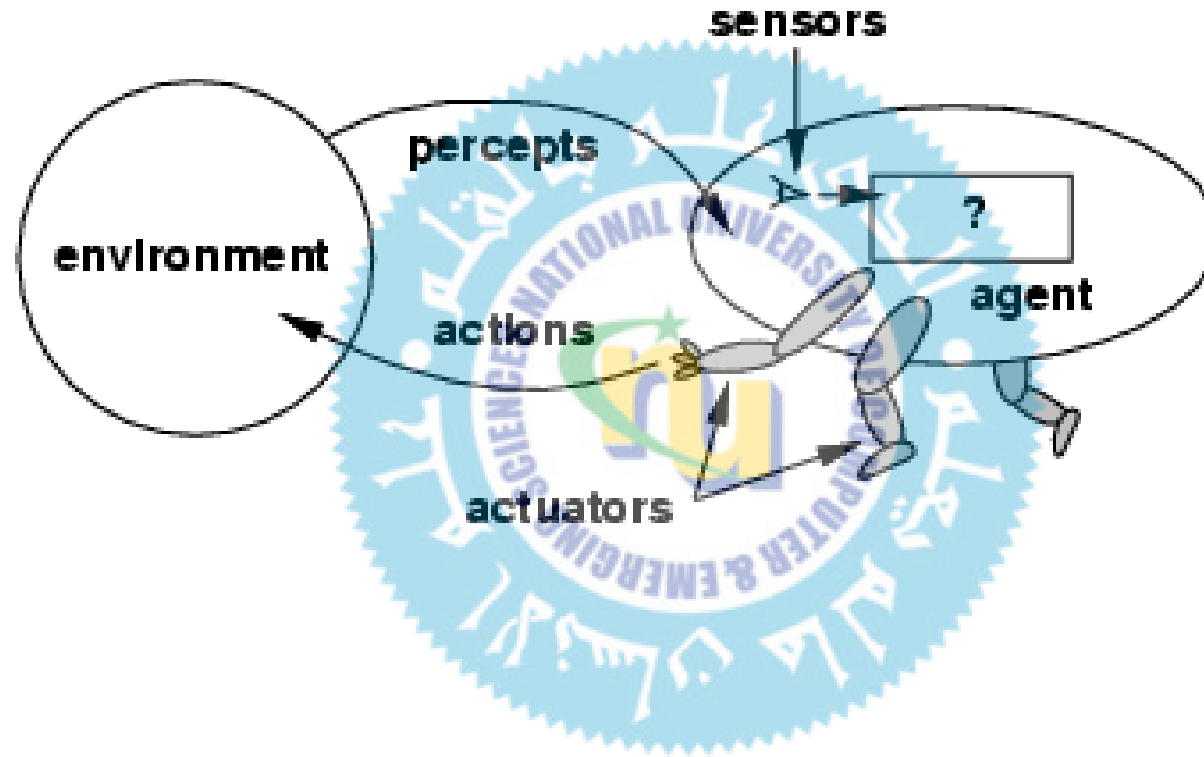
- "An agent is anything that can be viewed as **perceiving** its **environment** through sensors and **acting** upon that **environment** through effectors."



Russell & Norvig



Agent's function maps



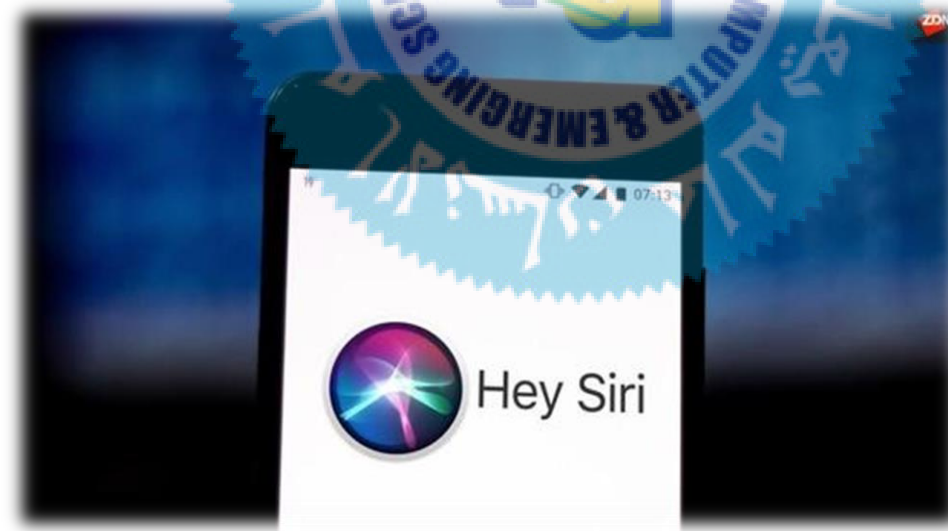
- The **agent function** maps from percept histories to actions:

Agents example

- **Human agent:**
 - eyes, ears, and other organs for sensors;
 - hands, legs, mouth, and other body parts for actuators
- **Robotic agent:**
 - cameras and infrared range finders for sensors
 - various motors for actuators

The Intelligent Agent

- AI assistants, *like Alexa and Siri, are examples of intelligent agents* as they use sensors to perceive a request made by the user and the automatically collect data from the internet without the user's help. They can be used to gather information about its perceived environment such as weather and time



The Intelligent Agent



```
Else  
If (Len(Trim$(str)  
  binFileNext =  
Else  
  Process data  
  intColon = InStr  
  (intColon  
  strName  
  strValue  
  strName =  
  Call xnNod  
End If  
End If  
End If
```



Humans

Programs

Robots

senses

keyboard, mouse, dataset

cameras, pads

body parts

monitor, speakers, files

motors, limbs

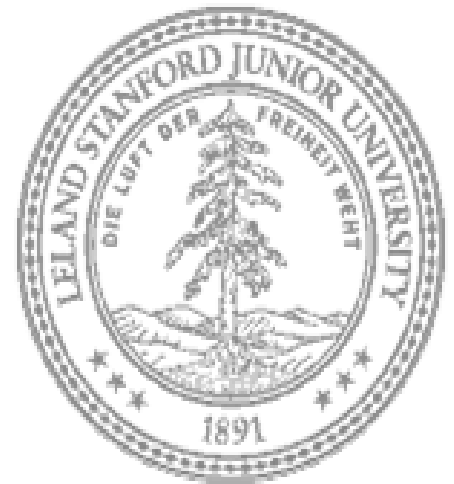
Agent Definition (1)

- An agent is an entity which is:
 - *Situated* in some *environment*.
 - *Autonomous*, in the sense that it can act *without direct intervention from humans* or other software processes, and controls over its own actions and internal state.
 - *Flexible* which means:
 - *Responsive (reactive)*: agents should *perceive their environment and respond to changes* that occur in it;
 - *Proactive*: agents should not simply act in response to their environment, they should be able *to exhibit opportunistic, goal-directed behavior* and take the initiative when appropriate;
 - *Social*: agents should be able *to interact with humans or other artificial agents*

Agent Definition (5)

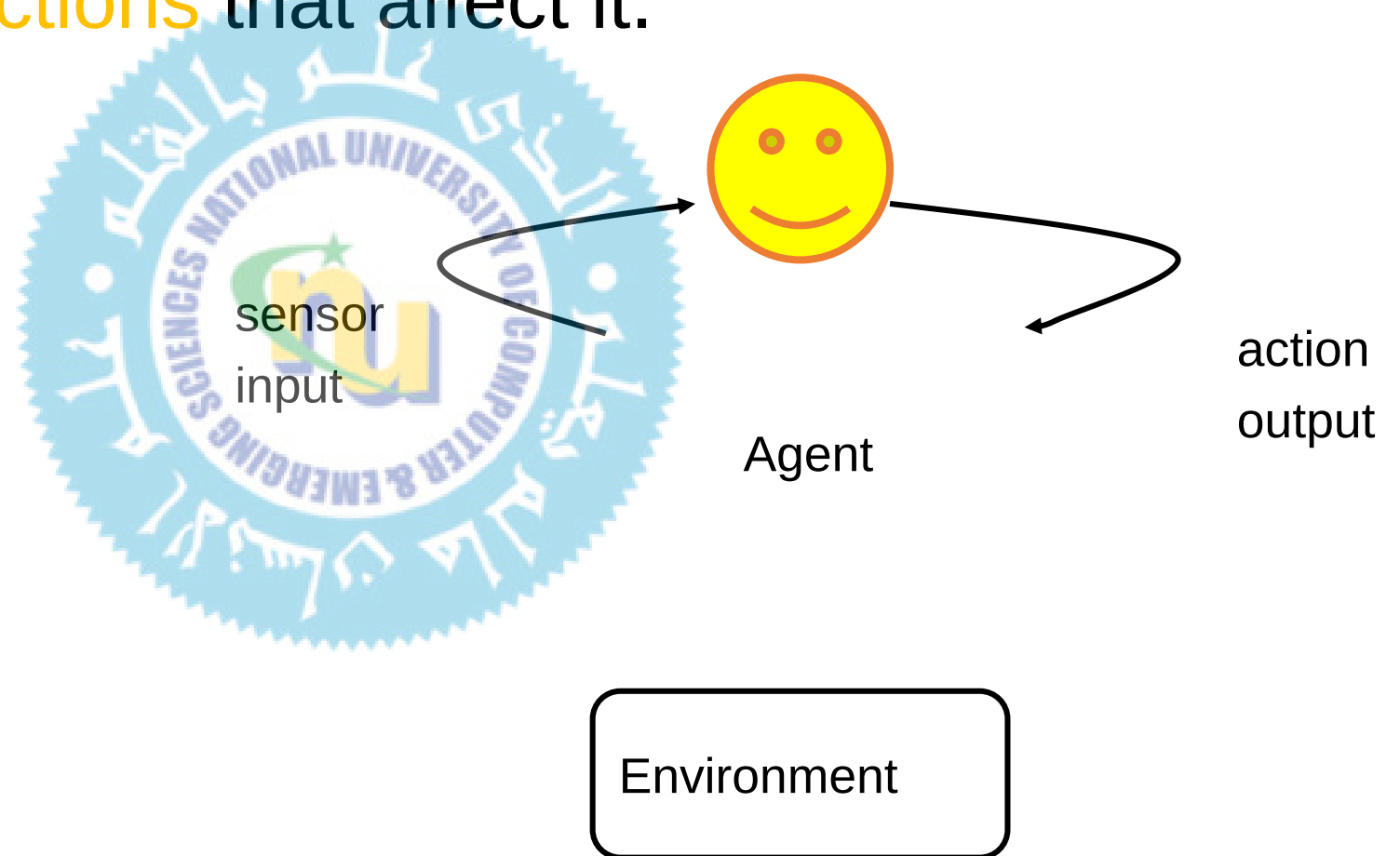
- “Intelligent agents continuously perform three functions: **perception** of dynamic conditions in the environment; **action** to affect conditions in the environment; and **reasoning** to interpret perceptions, solve problems, draw inferences, and determine actions.”

Barbara Hayes-Roth



Agents & Environments

- ❑ The agent takes **sensory input** from its **environment**, and produces as **output actions** that affect it.



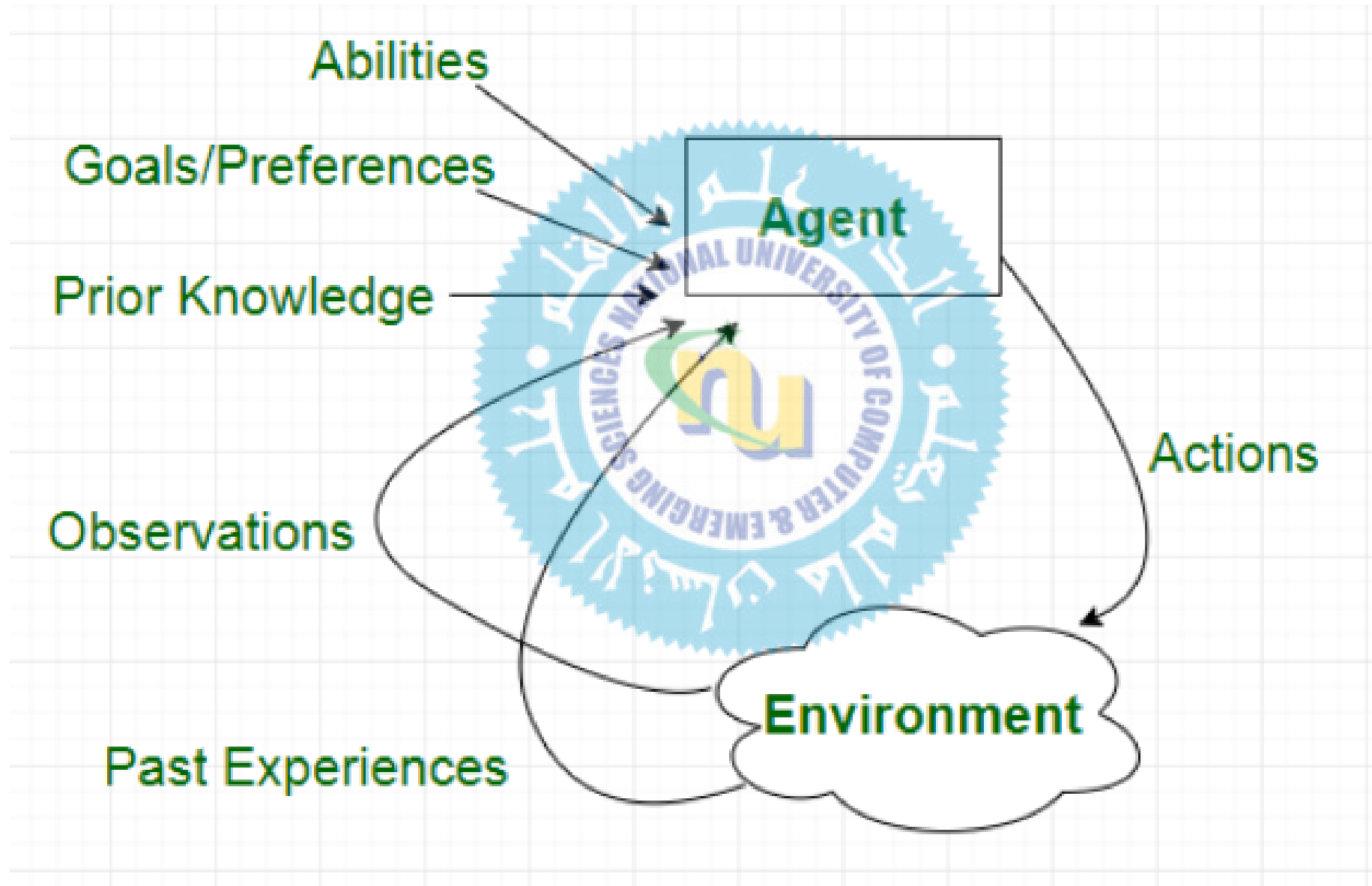
Examples of Agent

A **software agent** has *Keystrokes, file contents, received network packages which act as sensors and displays on the screen, files, sent network packets acting as actuators.*

A **Human agent** has eyes, ears, and other organs which act as sensors and hands, legs, mouth, and other body parts acting as actuators.

A **Robotic agent** has Cameras and infrared range finders which act as sensors and various motors acting as actuators.

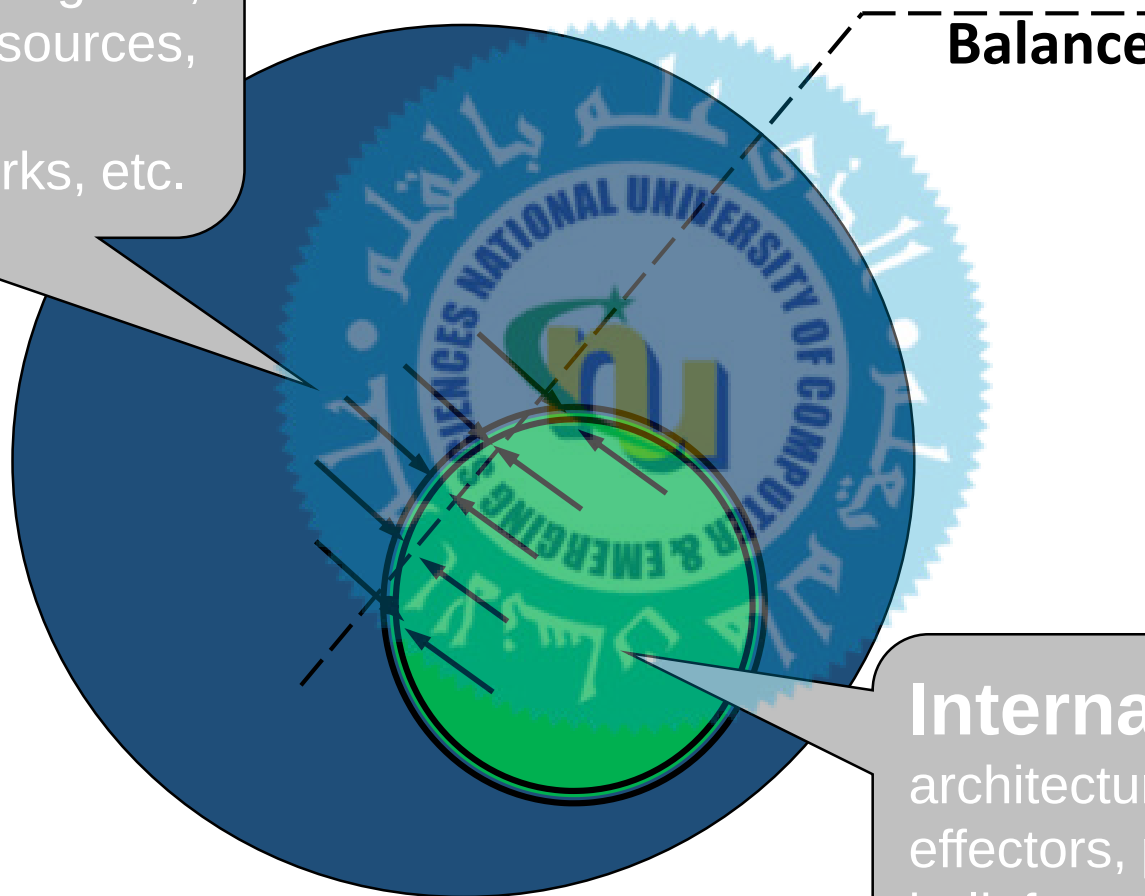
Characteristics of Agent



Internal and External Environment of an Agent

External Environment:

user, other humans, other agents, applications, information sources, their relationships, platforms, servers, networks, etc.



Balance

Internal Environment:

architecture, goals, abilities, sensors, effectors, profile, knowledge, beliefs, etc.

VACCUUM CLEANER AS AN AGENT

To illustrate these ideas, we use a very simple example—the vacuum-cleaner world shown in Figure 2.2. This world is so simple that we can describe everything that happens; it's also a made-up world, so we can invent many variations. This particular world has just two locations: squares *A* and *B*. The vacuum agent perceives which square it is in and whether there is dirt in the square. It can choose to move left, move right, suck up the dirt, or do nothing. One very simple agent function is the following: if the current square is dirty, then suck; otherwise, move to the other square. A partial tabulation of this agent function is shown in Figure 2.3 and an agent program that implements it appears in Figure 2.8 on page 48.

VACCUUM CLEANER AS AN AGENT

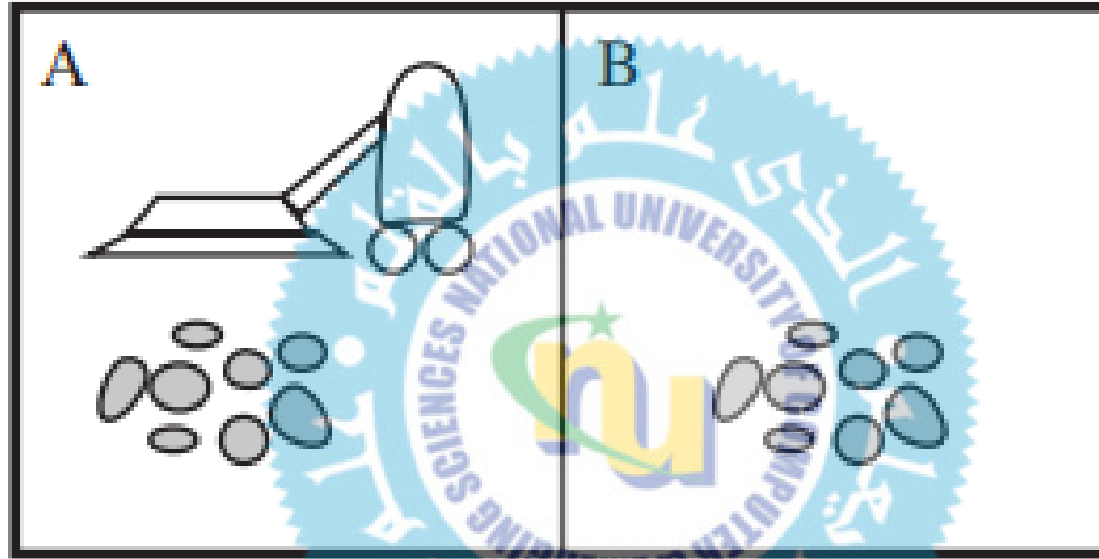


Figure 2.2 A vacuum-cleaner world with just two locations.

AGENT PERCEPTION SEQUENCE

Percept sequence	Action
<i>[A, Clean]</i>	<i>Right</i>
<i>[A, Dirty]</i>	<i>Suck</i>
<i>[B, Clean]</i>	<i>Left</i>
<i>[B, Dirty]</i>	<i>Suck</i>
<i>[A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>
<i>[A, Clean], [A, Clean], [A, Clean]</i>	<i>Right</i>
<i>[A, Clean], [A, Clean], [A, Dirty]</i>	<i>Suck</i>
<i>⋮</i>	<i>⋮</i>

Figure 2.3 Partial tabulation of a simple agent function for the vacuum-cleaner world shown in Figure 2.2.

Overall Intelligent Agent

- 1) is **goal-oriented**, because it should have at least one goal - *to keep continuously balance between its internal and external environments* ;
- 2) is **creative** because of the ability to *change external environment*;
- 3) is **adaptive** because of the ability to *change internal environment*;
- 4) is **mobile** because of the ability to *move to another place*;
- 5) is **social** because of the ability to *communicate to create a community*;
- 6) is **self-configurable** because of the ability to *protect “mental health”* by sensing only a “suitable” part of the environment.

[Terziyan, 1993]

Agent Definition

- ❑ In computer science, an **intelligent agent (IA)** is a software agent that exhibits some form of artificial intelligence that assists the user and will act on their behalf, in performing non-repetitive computer-related tasks.
- ❑ While the working of software agents used for *operator assistance* or data mining is often based on fixed pre-programmed rules, "intelligent" here implies the ability to adapt and learn.

Three groups of agents

- **Backseat driver**: helps the user during some task (e.g., Microsoft Office Assistant);
- **Taxi driver**: knows where to go when you tell the destination;
- **Caretaker** : know where to go, when and why.

What is Rationality & Rational Thinking

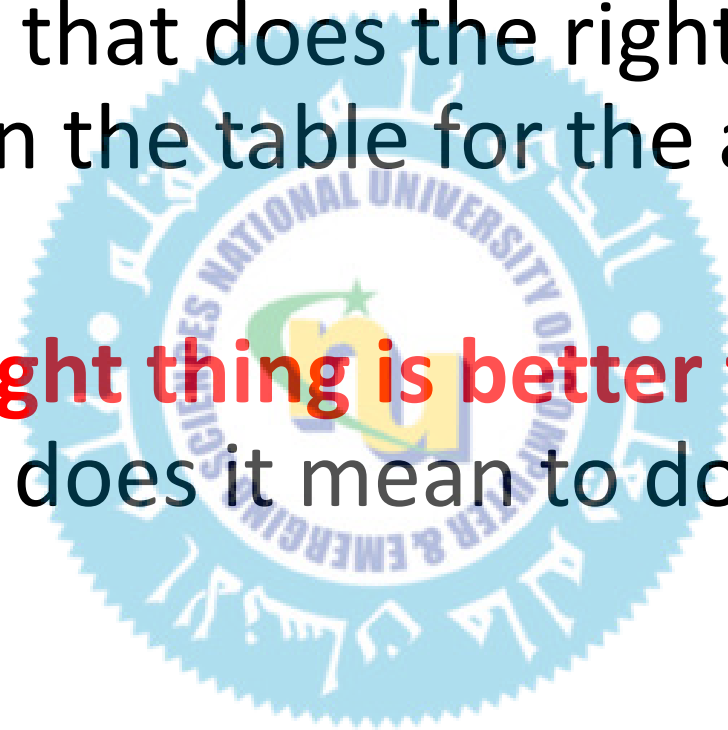
“Rationality is the quality or state of being rational – that is, being based on or agreeable to reason. Rationality implies the conformity of one's beliefs with one's reasons to believe, and of one's actions with one's reasons for action” (*Wikipedia*).

We humans have rational behavior by Birth

What is Rational Agent?

A **rational agent** is one that does the right thing—conceptually speaking, every entry in the table for the agent function is filled out correctly.

Obviously, **doing the right thing is better than doing the wrong thing**, but what does it mean to do the right thing?



So how to identify if agent is doing right things?

- We answer this age-old question in an age-old way: by considering the *consequences* of the agent's behavior.
- When an agent is plunked down in an environment, it generates a sequence of actions according to the percepts it receives.
- This sequence of actions causes the environment to go through a sequence of states.
- If the sequence is desirable, then the agent PERFORMANCE MEASURE has performed well.
- This notion of desirability is captured by a **performance measure** that evaluates any given sequence of environment states.

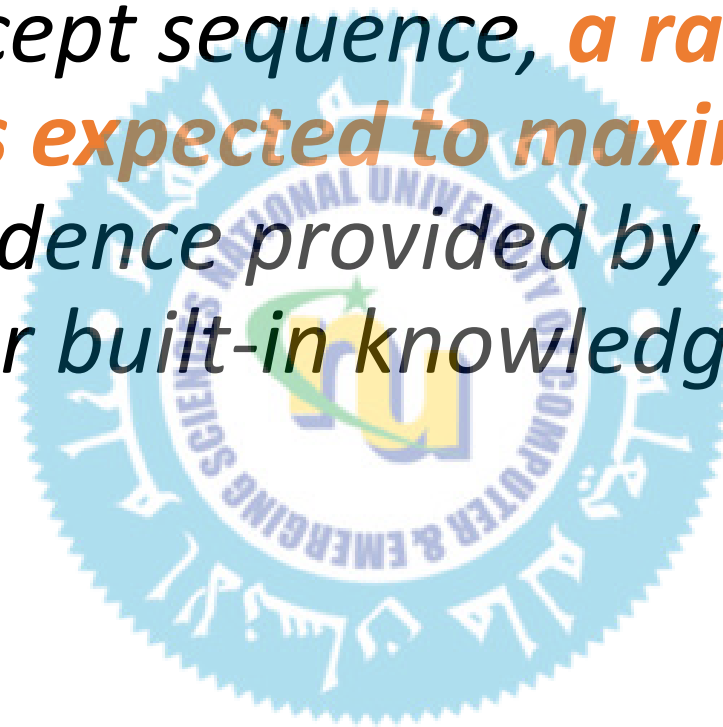
So how to identify if agent is doing right things?

*Generally speaking, it is better to **design performance measures** according to what one actually wants in the environment, rather than according to how one thinks the agent should behave.*



Rational Agent (Definition)

*“For each possible percept sequence, **a rational agent should select an action that is expected to maximize its performance measure**, given the evidence provided by the percept sequence and whatever **built-in knowledge** the agent has.”*

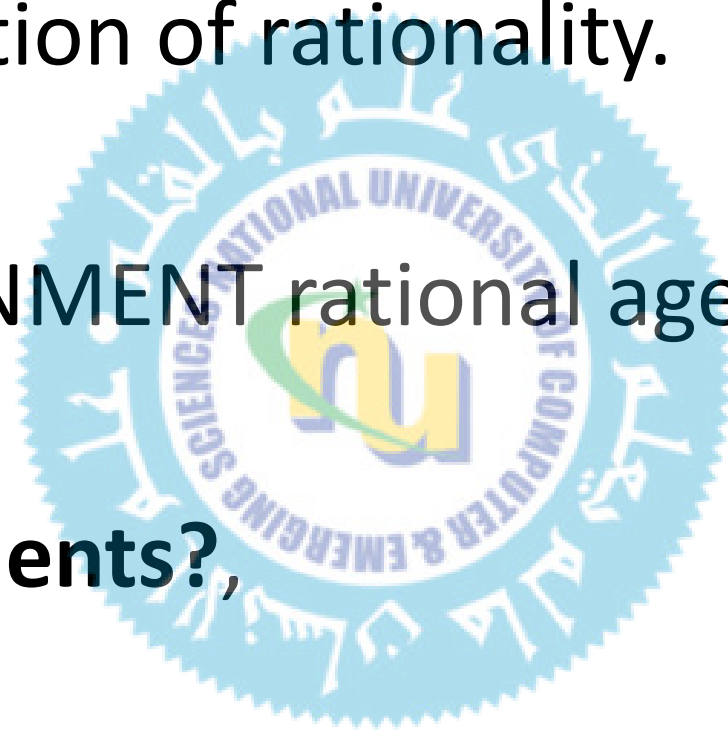


TASK ENVIRONMENT

Now, We have a definition of rationality.

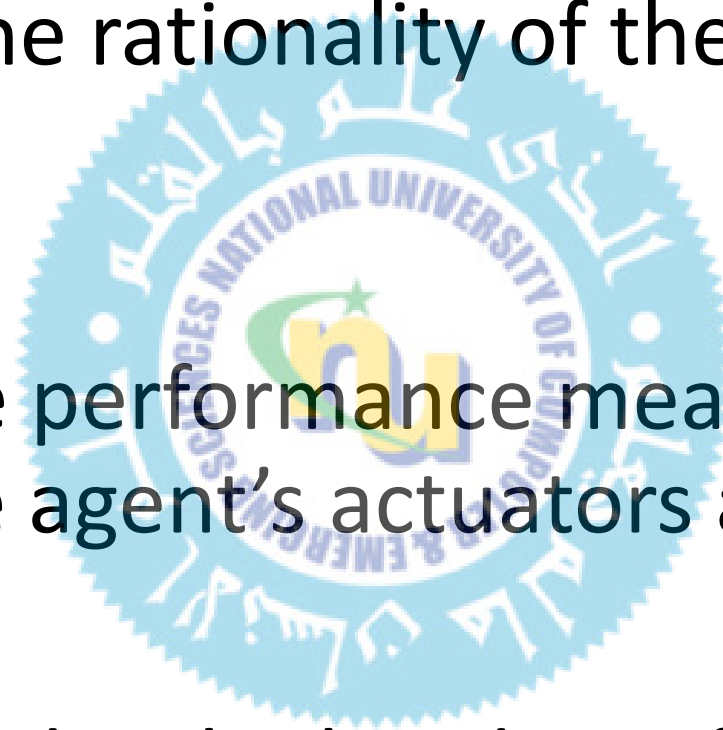
Building TASK ENVIRONMENT rational agents.

What is task environments?,



TASK ENVIRONMENT

- In our discussion of the rationality of the simple vacuum-cleaner agent,
- We had to specify the performance measure, the environment, and the agent's actuators and sensors.
- We group all these under the heading of the **task environment**. We call it PEAS.



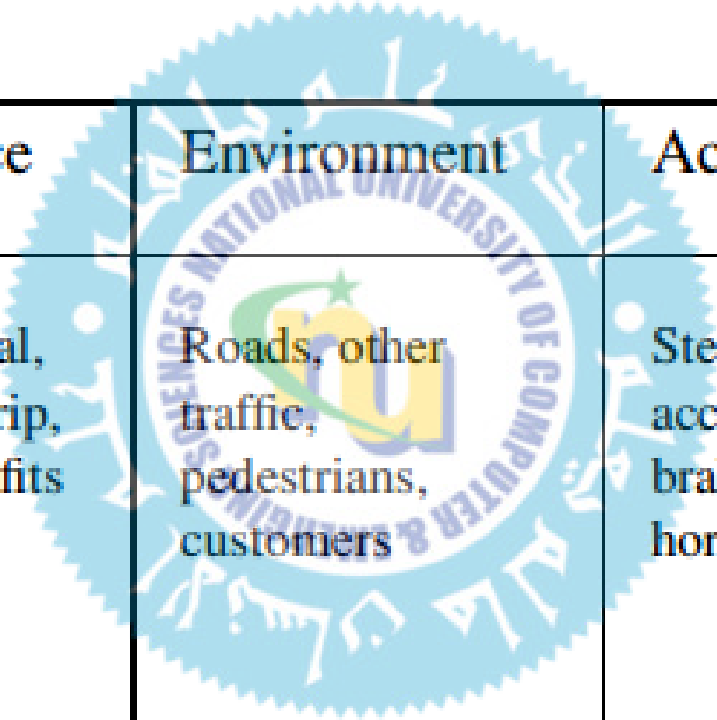
PEAS

- Use PEAS to describe task environment
 - Performance measure
 - Environment
 - Actuators
 - Sensors
- Example: Taxi driver
 - Performance measure: safe, fast, comfortable (maximize profits)
 - Environment: roads, other traffic, pedestrians, customers
 - Actuators: steering, accelerator, brake, signal, horn
 - Sensors: cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors



PEAS

- Example: Taxi driver



Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits	Roads, other traffic, pedestrians, customers	Steering, accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Figure 2.4 PEAS description of the task environment for an automated taxi.

PEAS

- Agent: Part-picking robot
- Performance measure: Percentage of parts in correct bins
- Environment: Conveyor belt with parts, bins
- Actuators: Jointed arm and hand
- Sensors: Camera, joint angle sensors

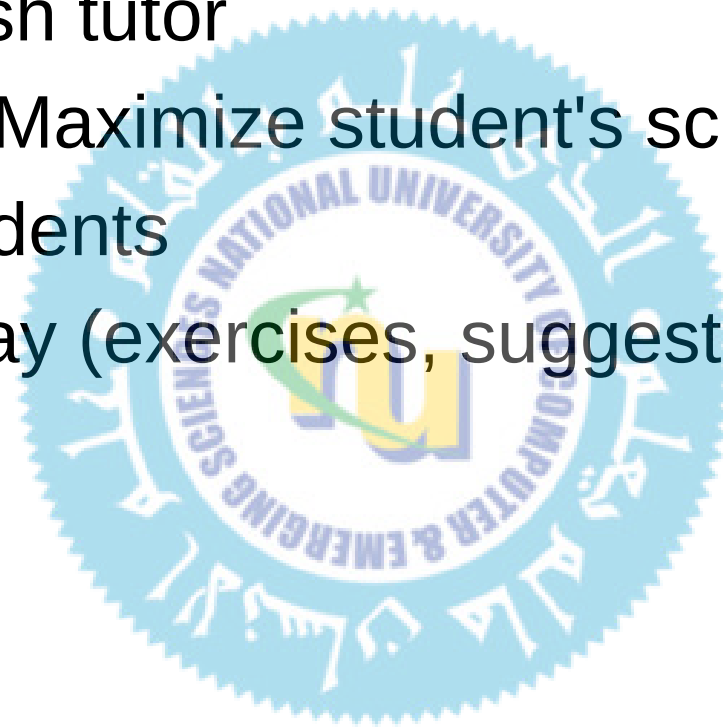


PEAS

- Agent: Medical diagnosis system
- Performance measure: Healthy patient, minimize costs, lawsuits
- Environment: Patient, hospital, staff
- Actuators: Screen display (questions, tests, diagnoses, treatments, referrals)
-
- Sensors: Keyboard (entry of symptoms, findings, patient's answers)

PEAS

- Agent: Interactive English tutor
- Performance measure: Maximize student's score on test
- Environment: Set of students
- Actuators: Screen display (exercises, suggestions, corrections)
- Sensors: Keyboard



Environment Types with real world examples

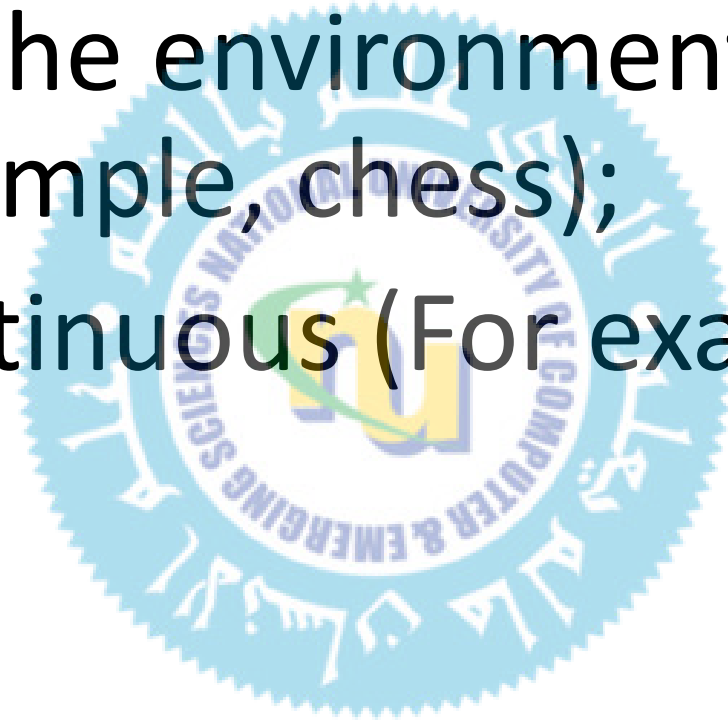
Fully observable vs. partially observable

If an agent's sensors give it access to the **FULLY OBSERVABLE** *complete state of the environment* at each point in time, then we say that the task environment is fully observable. For example

- A vacuum agent with only a local dirt sensor cannot tell whether there is dirt in other squares,
- An automated taxi cannot see *what other drivers are thinking*

Discrete / Continuous:

If there are a limited number of distinct, clearly defined, states of the environment, the environment is discrete (For example, chess); otherwise it is continuous (For example, driving).



Static / Dynamic:

If the environment does not change while an agent is acting, then it is static; otherwise it is dynamic.

E.g. – *Playing football, other players make it dynamic, mowing a lawn is static (unless there is a cat...), expert systems usually static (unless knowledge changes)*

Single agent / Multiple agents:

The environment may contain other agents which may be of the same or different kind as that of the agent.



Deterministic / Non-deterministic:

If the next state of the environment is completely determined by the current state and the actions of the agent, then the environment is deterministic; otherwise it is non-deterministic.

Examples:

Non-deterministic environment: physical world: Robot on Mars

Deterministic environment: Tic Tac Toe game

Episodic / Non-episodic:

- In an episodic environment, each agent's performance is the result of a series of independent tasks performed.
- If tasks to be performed are dependent then Non-Episodic

Episodic environment: mail sorting system, expert advice systems etc.

Non-episodic environment: chess game

Agent Structure

Agent consists of Agent Architecture and Agent Program.

Agent = Architecture + Agent Program



Agent Structure

Agent consists of Agent Architecture and Agent Program.

Architecture is the machinery that the agent executes on. Like a device with sensors and actuators, for example : a robotic car, a camera, a PC.

Agent program is an implementation of an agent function. An **agent function** is a map from the percept sequence(history of all that an agent has perceived till date) to an action.

Types of Agents

Agents can be grouped into four classes based on their degree of perceived intelligence and capability :

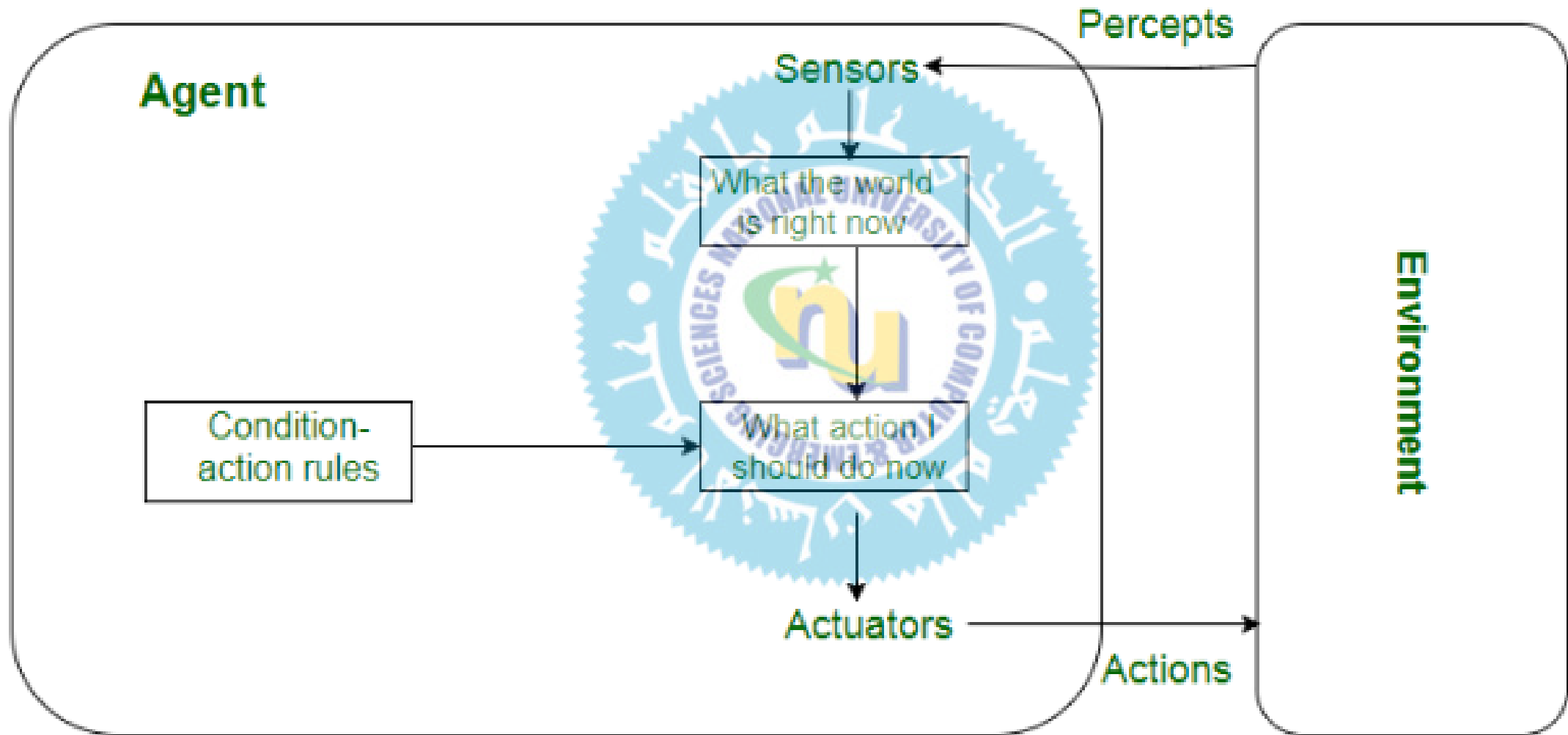
- Simple Reflex Agents
- Model-Based Reflex Agents
- Goal-Based Agents
- Utility-Based Agents



Simple reflex agents

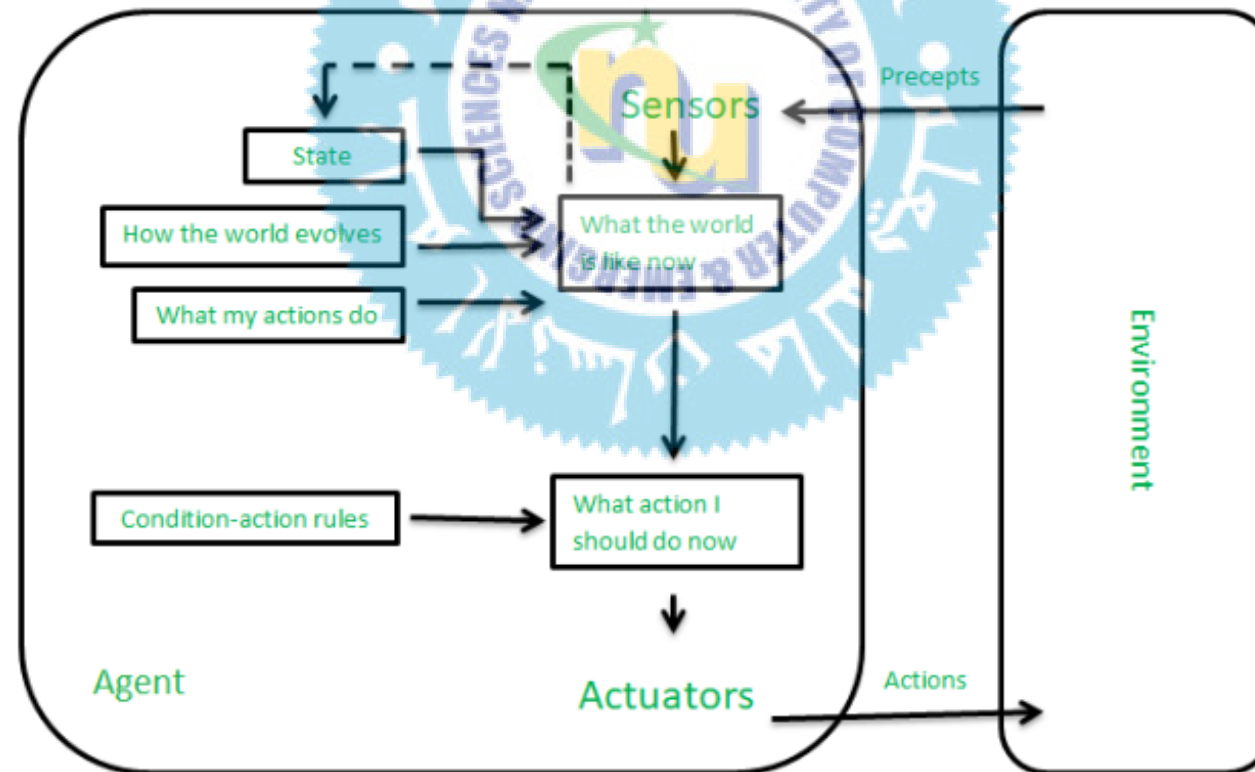
- Simple reflex agents ignore the rest of the percept history and act only on the basis of the ***current percept***.
- Percept history is the history of all that an agent has perceived till date.
- The agent function is based on the ***condition-action rule***.
- *Very limited intelligence.*
- *No knowledge of non-perceptual* parts of state.
- Usually too *big to generate* and store.

Simple reflex agents



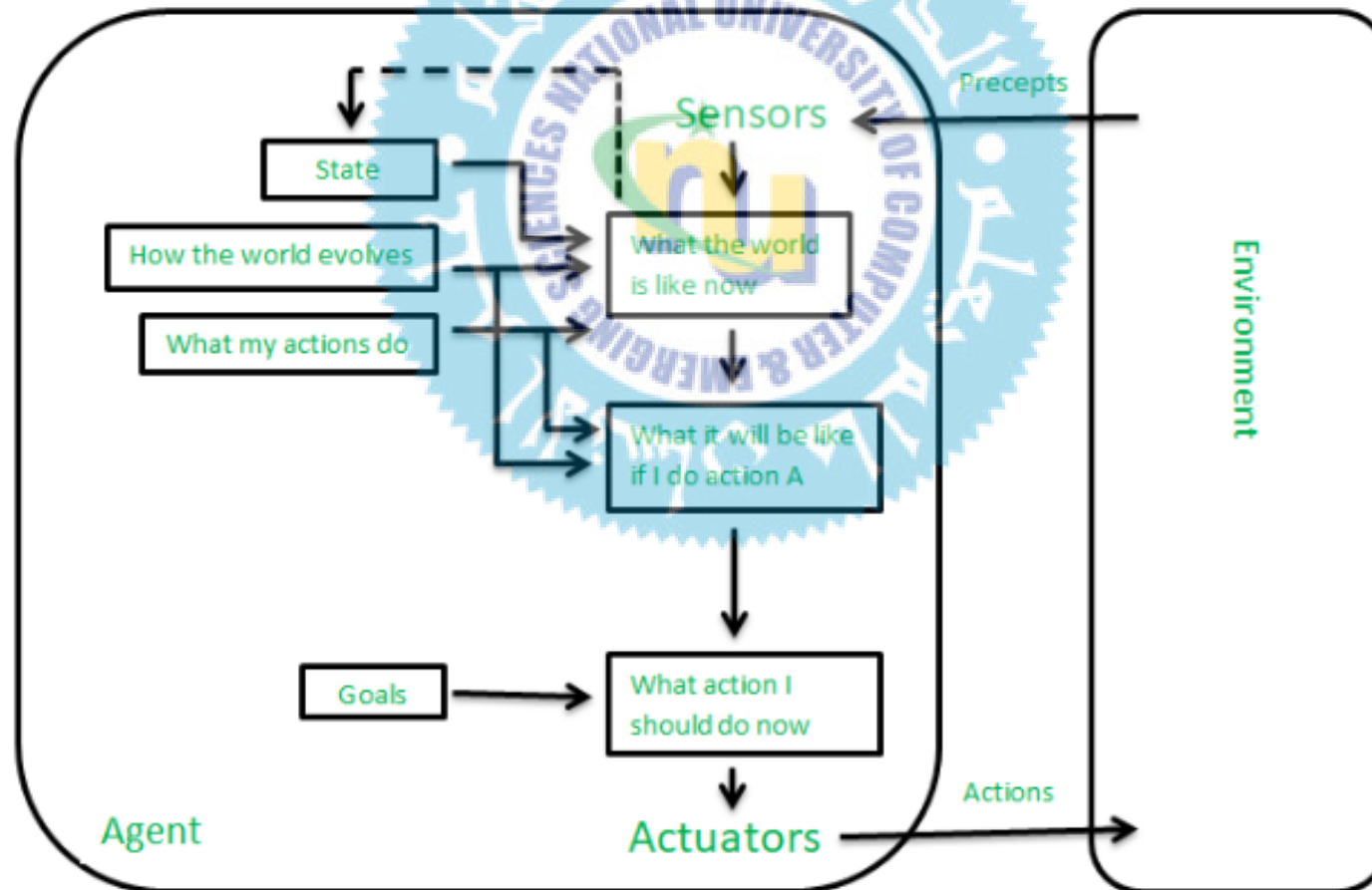
Model-based reflex agents

- A model-based agent can handle **partially observable environments** by use of model about the world.
- The agent has to keep track of **internal state** which is adjusted by each percept and that depends on the percept history.



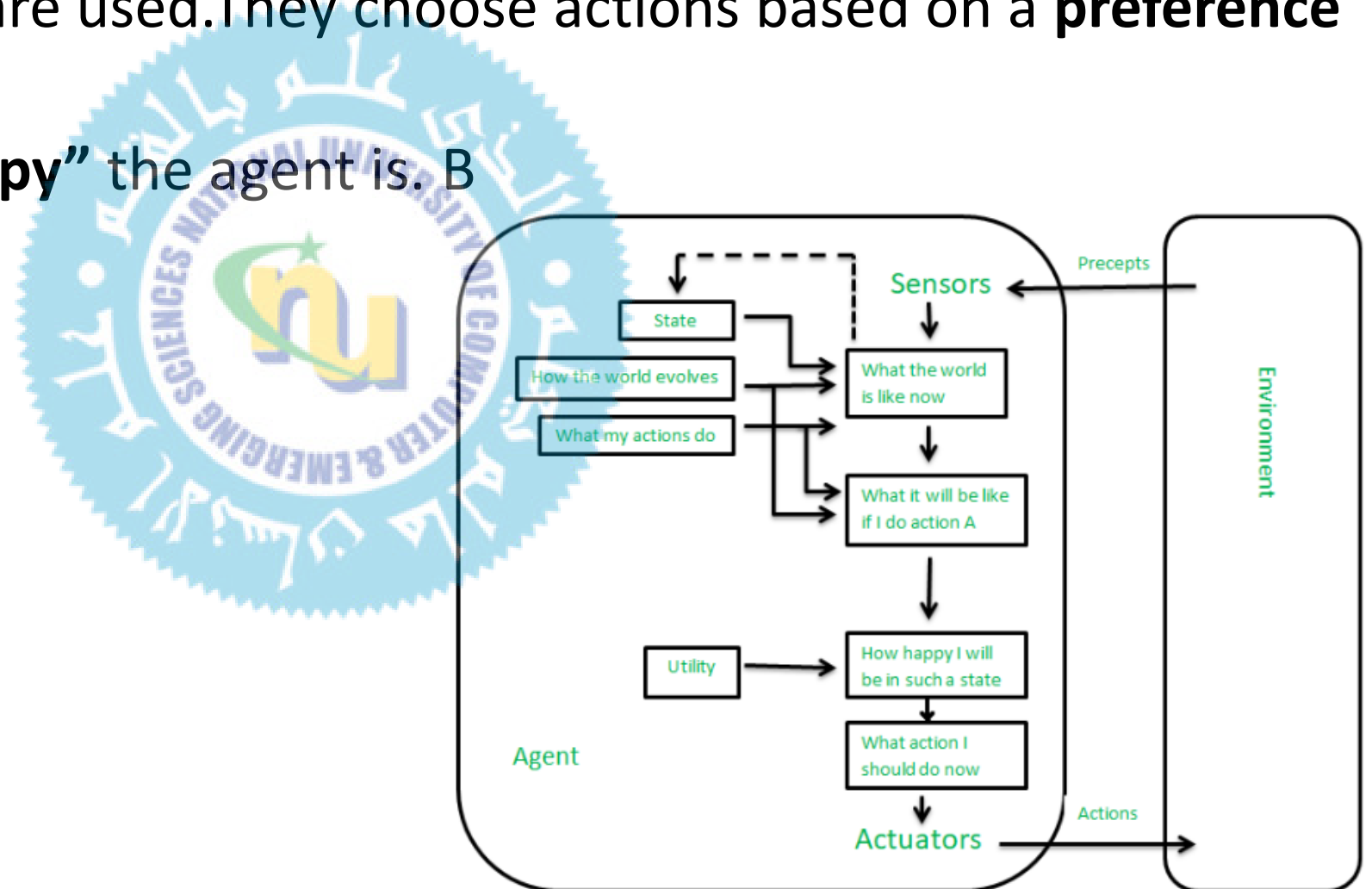
Goal-based agents

- These kind of agents take decision based on how far they are currently from their **goal**(description of desirable situations). Their every action is intended to reduce its distance from the goal.



Utility-based agents

- When there are multiple possible alternatives, then to decide which one is best, utility-based agents are used. They choose actions based on a **preference (utility)** for each state.
- Utility describes how “**happy**” the agent is. B

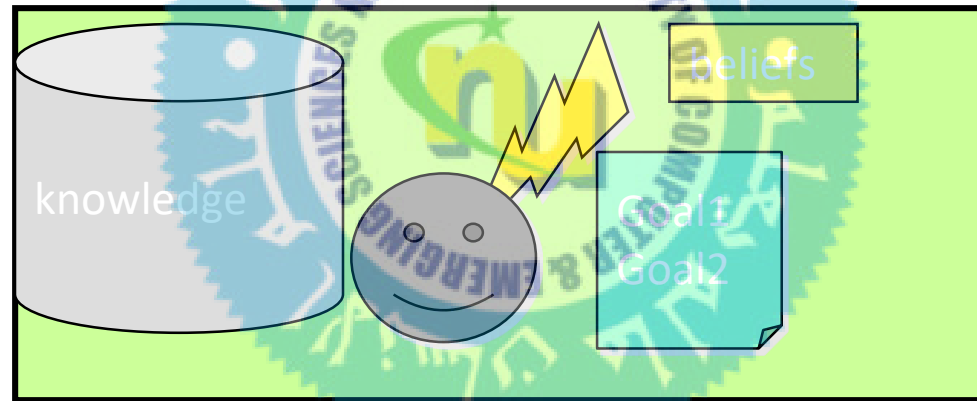


Agents and Their Actions

- a rational agent does “the right thing”
 - the action that leads to the best outcome under the given circumstances
- an agent function maps percept sequences to actions
 - abstract mathematical description
- an agent program is a concrete implementation of the respective function
 - it runs on a specific agent architecture (“platform”)

Description of Agent's character

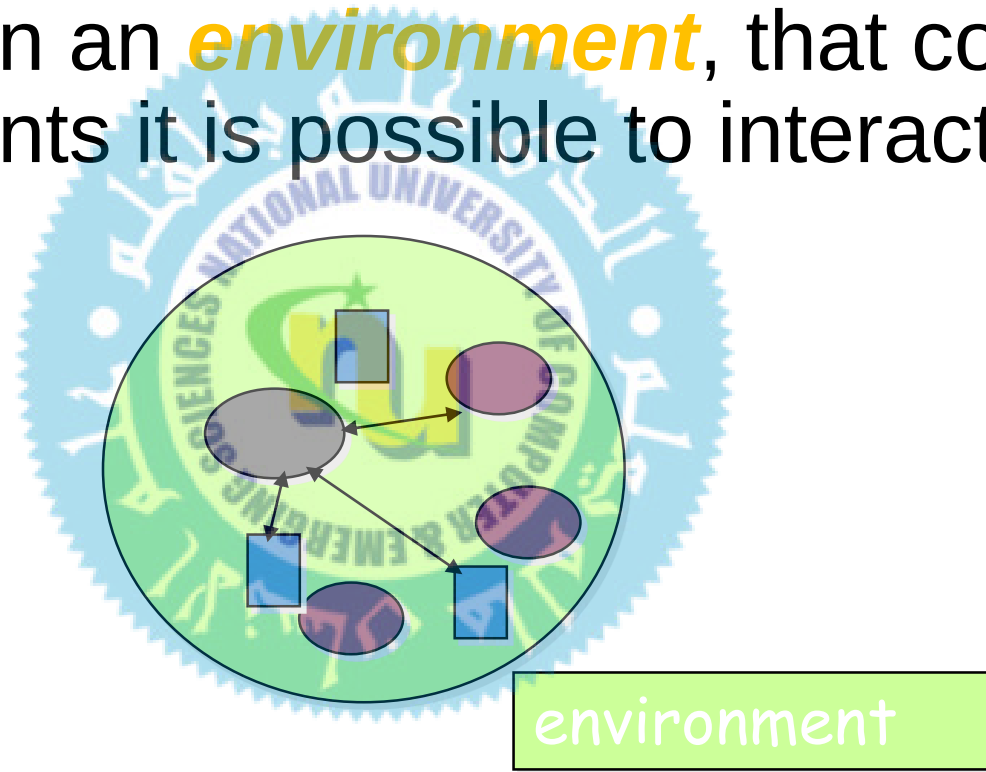
- ❑ An agent is responsible for satisfying specific *goals*. There can be different types of goals such as achieving a specific status, maximising a given function (e.g., utility), etc.



- ❑ The **state** of an agent includes state of its **internal environment** + **state of knowledge** and **beliefs** about its **external environment**.

Connections

- An agent is *situated* in an *environment*, that consists of the objects and other agents it is possible to interact with.



- An agent has an *identity* that distinguishes it from the other agents of its environment.

Gift Shop Agent Example

Gift shop

Items such as caps, t-shirts, sweatshirts and other miscellanea such as buttons and mouse pads have been designed. In addition, merchandise for almost all of the projects is available.



Hi. I'm your automated online assistant. How may I help you?

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Gift Shop Agent Example

DID YOU EVER
INTERACT WITH AN
AGENT?

A large, light blue, semi-transparent watermark of the Beihang University logo is centered in the background. The logo is circular with a serrated edge and contains the university's name in English and Chinese, along with a stylized graphic of a book and a torch.

Summery of Today's Lecture

- What is an Intelligent agent?
- Agents & Environments
- Performance measure
- Environment
- Actuators
- Sensors
- Features of intelligent agents

